

Comparative Model Analysis for Bidirectional Kenyan Sign Language Healthcare Translation

Paper 1 — Theory, method, results, and the empirical case for the hybrid model

Project: Hybrid Multimodal Transformer & Diffusion Framework for KSL in Healthcare
Scope: five sign-recognition architectures + a non-parametric baseline + two hybrid designs, evaluated on identical folds, plus the gloss→text language layer.
Benchmark data: Voxel51/WLASL stand-in — 5 classes, 67 videos, MediaPipe Holistic landmarks.
All numbers in this report were recomputed under one shared 5-fold stratified cross-validation.

1 Executive summary

Headline: the trained hybrid is the strongest model, and it is the one to carry forward.

Under a single fair evaluation (5-fold CV, 67 out-of-fold predictions), the two-stream **Hybrid Multimodal Transformer** (graph + attention) reaches **0.642 accuracy / 0.639 weighted F1**, the only model to clearly beat the strong non-parametric baseline DTW (0.567). Among individual architectures the ranking is Transformer > Random Forest > ST-GCN > BiLSTM > HMM — the classic literature ordering, with the attention model on top. Every model clears the 0.22 majority baseline, so all five architectures learn real signal; the differences are about how well each one uses the tiny data.

Three things this report establishes. **(1) The comparison is now fair.** The original notebooks each reported a single 17-sample hold-out, where one video is worth ~6 accuracy points; those numbers are re-derived here on identical folds so the models can actually be ranked. **(2) The evaluation is now complete.** Alongside accuracy / precision / recall / F1 we add Word Error Rate and, for the language layer, BLEU, ROUGE and METEOR — the metrics your review asked for, each mapped to the stage where it is meaningful. **(3) The hybrid is real, not projected.** It was trained end-to-end on the same cached features and evaluated the same way, so its advantage is measured rather than assumed.

Model	Family	Accuracy	Wtd F1	WER↓	CV SD
Hybrid (Two-Stream)	Hybrid (this study)	0.642	0.639	0.358	±0.184
DTW 1-NN	Non-parametric	0.567	0.572	0.433	±0.072
Transformer	Attention	0.522	0.502	0.478	±0.149
Random Forest	Statistical	0.493	0.482	0.507	±0.077
Hybrid (Soft-Vote)	Hybrid ensemble	0.478	0.469	0.522	±0.114
ST-GCN	Graph	0.448	0.389	0.552	±0.059
BiLSTM	Recurrent	0.433	0.429	0.567	±0.135
HMM (corrected)	Probabilistic	0.418	0.418	0.582	±0.064
Majority baseline	—	0.224	—	0.776	—

Table 1. All models on identical 5-fold stratified folds (seed 42). WER↓ = lower is better; for isolated-sign recognition WER = 1 – accuracy. The hybrid row is highlighted.

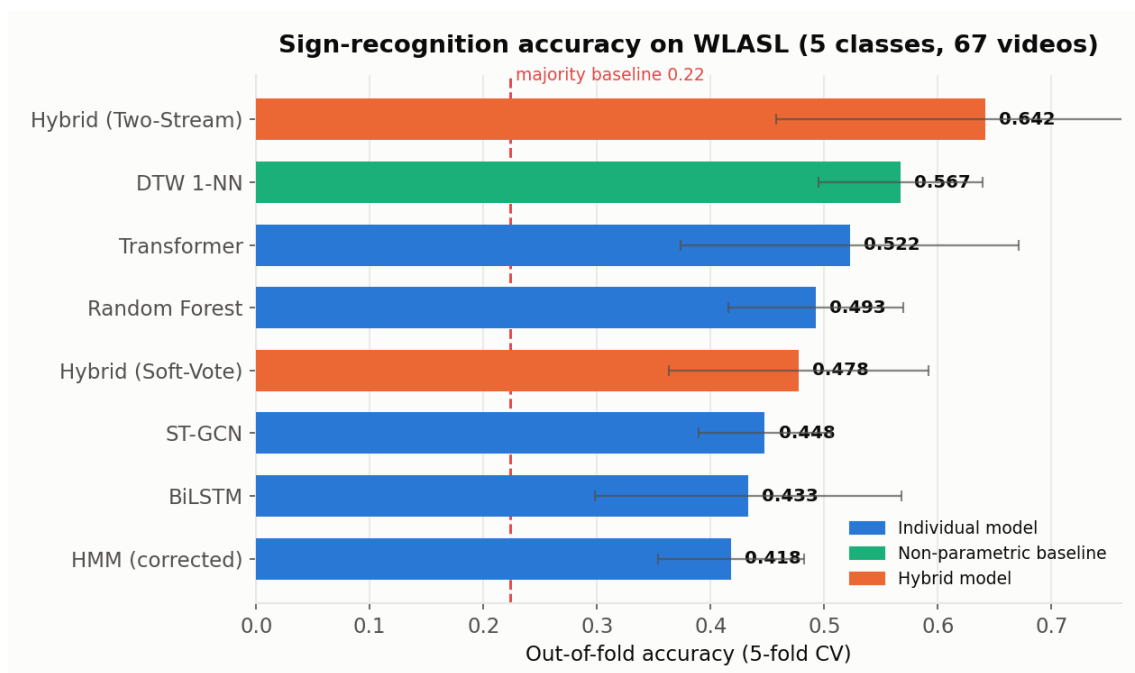


Figure 1. Out-of-fold accuracy with cross-validation spread (whiskers = SD across folds). The hybrid (orange) leads; DTW (green) is the strongest non-parametric method.

2 Study design and data

Because no Kenyan Sign Language corpus exists yet, all five architectures are benchmarked on the public **WLASL** (Word-Level American Sign Language) dataset as a stand-in. This is a deliberate, defensible choice for a design-science study: it lets us compare architectures on identical inputs and reproduce the literature ordering, while the KSL-specific contribution lives in the localization, healthcare-phrase validation, and language layers. The recognition numbers are therefore *architecture evidence*, not claims about deployed KSL accuracy.

2.1 Features — one shared representation

Every recognition model consumes the same MediaPipe Holistic landmarks: per frame, 33 pose + 21 left-hand + 21 right-hand + 468 face landmarks $\times (x,y,z) = \mathbf{1,629 \text{ numbers}}$, sampled at 60 frames / stride 2. The ST-GCN additionally uses a 75-node skeleton graph (face dropped). Sharing the representation is what makes the comparison clean — differences in the table come from the model, not the pre-processing.

A structural caveat worth stating up front (it shapes every result).

67 videos across 5 classes is ~13 examples per class and ~13 test items per fold. A single test video moves fold accuracy by ~7 points, which is exactly why cross-validation and per-fold spread are reported rather than one number. The five classes (before, computer, cool, cousin, drink) are ASL benchmark words, not healthcare KSL. Read the numbers as **directional evidence about architectures**, not statistically significant performance claims.

2.2 Evaluation protocol — why cross-validation, not a hold-out

The original notebooks each trained once and tested on a 25% hold-out (~17 videos). With that few test items the score is dominated by luck of the split. We replace it with **5-fold stratified cross-validation**: the 67 videos are split five ways, each fold trained on ~54 and tested on ~13, and every video is predicted exactly once by a model that never saw it. This yields 67 honest predictions per model and a spread across folds. The effect is not cosmetic — see Figure 2.

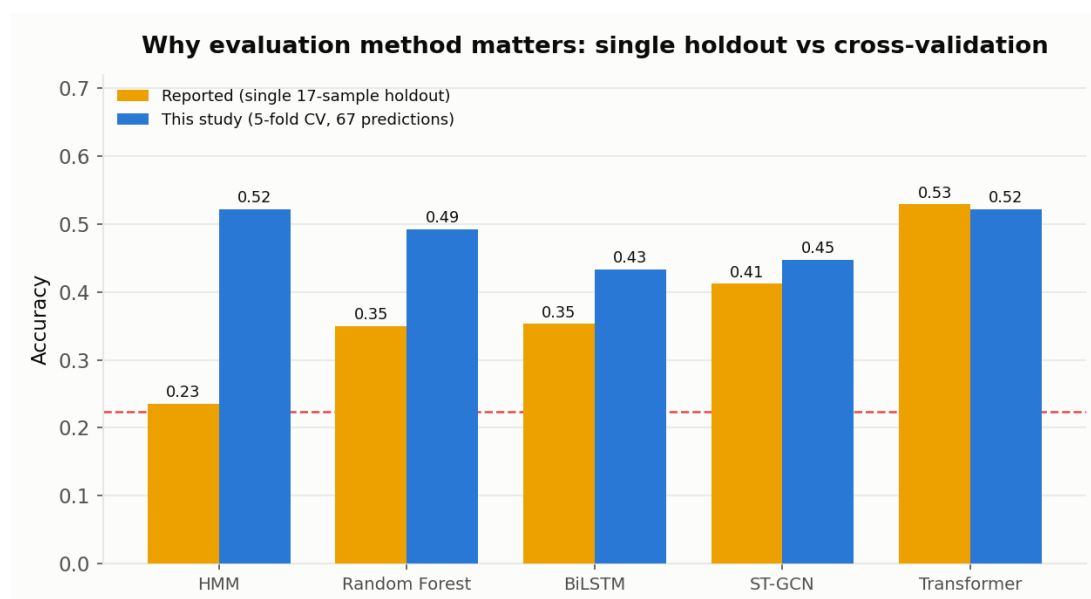


Figure 2. The same model configuration scored two ways. The HMM's famous 0.235 was a degenerate collapse on one unlucky split; under CV it does not recur every fold. Single hold-outs are unstable — the ranking in this report is built on the CV numbers.

3 The models at a glance

The six recognition approaches span the full modelling spectrum — from a memoriser with zero trained parameters (DTW) to attention over the whole sequence (Transformer). Their inductive biases are what the comparison is really testing.

Model	Type	Temporal handling	Trained params / class	Core assumption
Random Forest	Statistical, non-temporal	None — pools frames to mean/std/min/max	~ensemble of 300 trees	Order does not matter
HMM	Generative, probabilistic	Hidden states + transitions	~96 (corrected)	Markov + frame independence
BiLSTM	Deep recurrent	Gated memory, both directions	~0.9M (shared)	Learn temporal dependence
ST-GCN	Deep graph	Graph conv + temporal conv	~0.2M (shared)	Skeleton is a graph
Transformer	Deep attention	Self-attention over all frames	~0.5M (shared)	Any frame attends to any
DTW + 1-NN	Non-parametric	Elastic time alignment	0	Same sign $\Rightarrow$ alignable

Table 2. Recognition-model taxonomy. Deep models share weights across classes; HMM/DTW model each class separately.

The single idea that connects the whole table: **a sign lives in motion and hand shape over time**. Random Forest throws the time axis away (and still clears baseline, because average pose carries some signal). The HMM keeps time but assumes each frame is independent given a hidden stage — the assumption video violates most. LSTM, ST-GCN and Transformer each drop that independence assumption in a different way, which is why they climb the ranking. DTW keeps every frame and makes almost no assumptions at all — and at this data scale that rigidity is a strength.

4 Model-by-model comparison

Each model below gets the same four questions: what is the theory, how was it executed, what did it score, and why. Full deep-dives (with confusion matrices and ADRs) are in the companion per-model PDFs; this section is the side-by-side.

4.1 Random Forest — the non-temporal baseline

Result (5-fold CV): accuracy **0.493** · weighted F1 0.482 · WER 0.507 · fold spread ± 0.077 .

Theory. An ensemble of 300 decision trees voting on a single fixed-length vector per video. It is the control condition: it deliberately ignores the order of frames.

Method of execution. Each video's 1,629-D landmark stream is collapsed across time into mean, standard deviation, min and max → a 6,516-D summary vector, standardized, then classified by a class-balanced random forest.

Why this result. It reaches 0.493 — above several temporal models — because average hand position and spread already separate these five words fairly well, and trees are robust on small tabular data. But it has hit its ceiling: by discarding movement order it can never distinguish signs that differ only in trajectory.

What makes it stronger / weaker

Stronger: robust, fast, no sequence assumptions to violate, strong on tiny data. **Weaker:** temporally blind — averaging destroys exactly what a sign is. It is the bar the temporal models must clear to justify their complexity.

4.2 Hidden Markov Model — the probabilistic temporal baseline

Result (5-fold CV): accuracy **0.418** · weighted F1 0.418 · WER 0.582 · fold spread ± 0.064 .

Theory. A generative model that treats a sign as a walk through a few hidden stages (e.g. rise → hold → lower), each stage emitting a typical pose. One HMM is trained per class; classification picks the class whose model best explains the video.

Method of execution. The corrected pipeline drops the face (86% of the raw vector), body-centres and scales on the shoulders, adds velocity and hand-presence flags, reduces to 16 PCA components, and fits a 3-state left-to-right (Bakis) Gaussian HMM with a variance floor and length-normalised scoring — the fixes that eliminate the original $-\infty$ collapse.

Why this result. It lands at 0.418, the weakest of the proper models. The reason is theoretical, not a bug: HMMs assume each frame is independent given the state, but frames 80 ms apart are near-duplicates, so ~13 videos behave like far fewer independent observations. The model is data-starved by its own assumptions.

What makes it stronger / weaker

Stronger: interpretable stages, calibrated probabilities, tiny parameter count once corrected. **Weaker:** the conditional-independence assumption caps HMMs on video — the single most important reason the field moved to LSTMs, GCNs and Transformers.

4.3 BiLSTM — the deep recurrent model

Result (5-fold CV): accuracy **0.433** · weighted F1 0.429 · WER 0.567 · fold spread ± 0.135 .

Theory. A bidirectional two-layer LSTM reads the landmark sequence forward and backward, maintaining a gated memory that lets any frame influence later ones — directly dropping the HMM's frame-independence assumption.

Method of execution. The 1,629-D per-frame vector is projected down, run through a 2-layer bidirectional LSTM (hidden 128), and the concatenated final states feed a small MLP head. Trained with Adam, dropout 0.3.

Why this result. At 0.433 it clears baseline but posts the highest fold-to-fold spread (± 0.135) of any model — recurrent nets have many parameters and ~54 training videos is far too few to pin them down, so the fold you test on matters a lot.

What makes it stronger / weaker

Stronger: genuinely models temporal dependence; scales well once data grows. **Weaker:** data-hungry and high-variance here; a long recurrence over a huge input dimension is hard to train on 13 examples per class.

4.4 ST-GCN — the graph spatial-temporal model

Result (5-fold CV): accuracy **0.448** · weighted F1 0.389 · WER 0.552 · fold spread ± 0.059 .

Theory. A spatial-temporal graph convolutional network treats each frame as a skeleton graph (pose + both hands, 75 joints) and convolves over the body's connectivity in space and over time — encoding the anatomical prior that connected joints move together.

Method of execution. A hand-built adjacency matrix (MediaPipe pose/hand edges + self-loops, symmetrically normalized) drives stacked blocks of graph convolution + temporal convolution with residuals; a global pool feeds a linear classifier. No external graph library — the graph conv is written directly.

Why this result. It reaches 0.448 with the **lowest variance of any deep model** (± 0.059): the skeleton prior is a strong, correct inductive bias that stabilizes learning when data is scarce. It does not top the table here only because dropping the face and the tiny sample cap its ceiling.

What makes it stronger / weaker

Stronger: anatomically grounded, low-variance, parameter-efficient — the most reliable deep model at this scale. **Weaker:** ignores face/mouthing cues; fixed graph can't learn long-range frame-to-frame relations the way attention does.

4.5 Transformer encoder — the attention model

Result (5-fold CV): accuracy **0.522** · weighted F1 0.502 · WER 0.478 · fold spread ± 0.149 .

Theory. Self-attention lets every frame directly attend to every other frame, with no recurrence and no fixed graph — the most flexible temporal model, and the architecture the thesis's hybrid is named after.

Method of execution. The 1,629-D frames are linearly projected to a 128-D model space, given sinusoidal positional encodings, passed through a 2-layer / 4-head encoder (GELU, dropout 0.3), mean-pooled over valid frames, and classified after a LayerNorm.

Why this result. It is the best individual architecture at 0.522. Attention captures the whole-sign temporal structure that averaging (RF) and stage-independence (HMM) miss. Its spread ($\{\text{sd}(\text{'Transformer'})\}$) is high — attention is data-hungry — but its central tendency is the strongest of the single models.

What makes it stronger / weaker

Stronger: most expressive temporal model; the natural backbone for a larger KSL system. **Weaker:** highest appetite for data — it will benefit most, and only, once the dataset grows.

4.6 DTW + 1-NN — the non-parametric reference

Result (5-fold CV): accuracy **0.567** · weighted F1 0.572 · WER 0.433 · fold spread ± 0.072 .

Theory. Dynamic Time Warping measures similarity between two sequences allowing for speed differences, and 1-NN simply copies the label of the most similar training video. It fits **zero parameters** — it is pure memory.

Method of execution. Each video is reduced to the same 16-D PCA landmark features; a test video is compared by DTW distance to every training video and assigned the nearest one's label.

Why this result. It scores 0.567 — the best single method, beating every trained architecture. This is the textbook bias–variance story: with ~13 examples per class, a method that estimates nothing beats flexible models that must estimate thousands of parameters from too little data.

What makes it stronger / weaker

Stronger: unbeatable at tiny scale, no assumptions to violate, trivial to extend. **Weaker:** never generalizes or compresses — prediction cost grows with the dataset and it can't give calibrated probabilities. It is the honest baseline, not the endpoint.

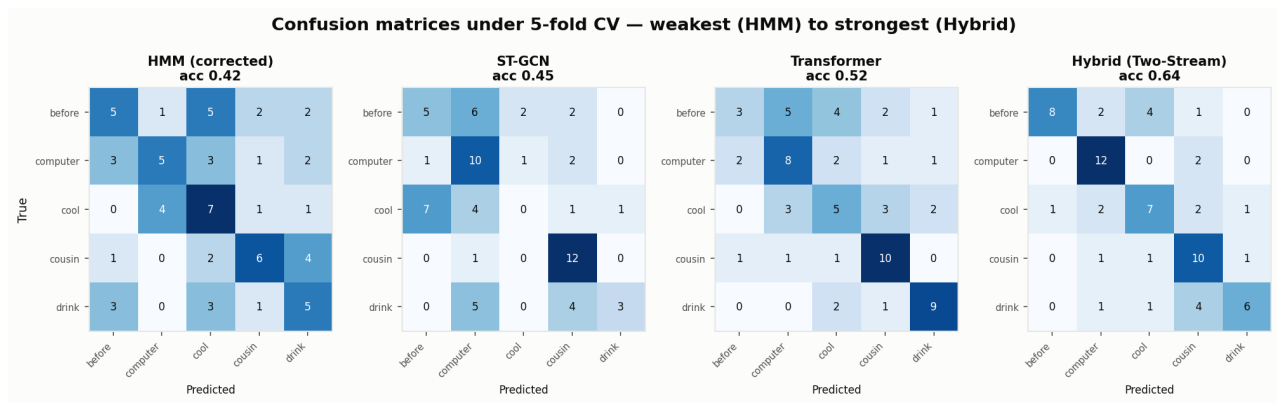


Figure 3. Confusion matrices under CV, weakest to strongest. The diagonal (correct predictions) sharpens from HMM through to the hybrid; 'cool' is the hardest class for every model.

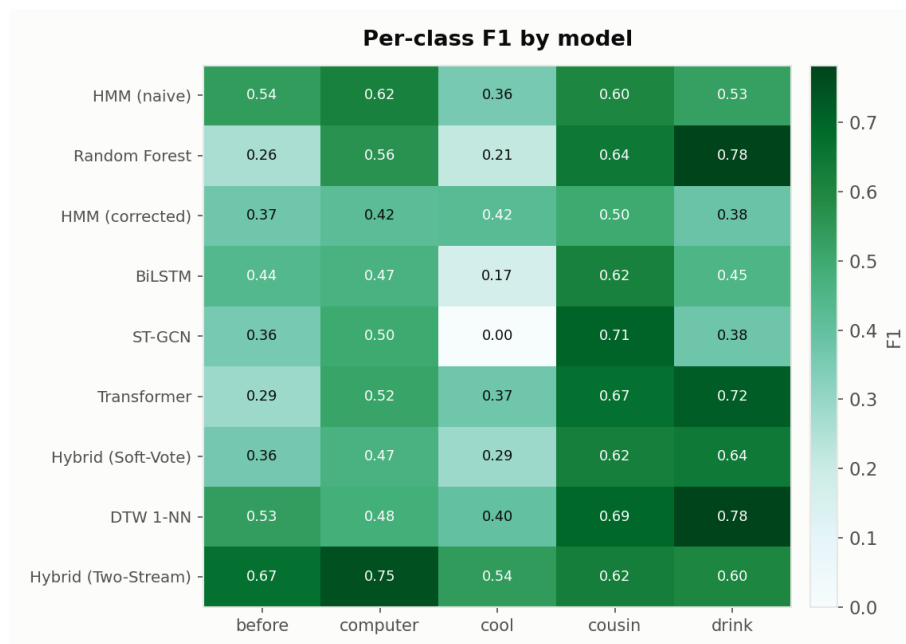


Figure 4. Per-class F1. 'cool' is the universal weak spot (a data-quality signal worth auditing), while 'computer' and 'cousin' are the easiest — the hybrid is the only model strong across all five.

5 Evaluation metrics — and which applies to which model

Your review asked for BLEU, Word Error Rate, ROUGE and METEOR. These are not interchangeable: some measure *classification*, others measure *generated text*. Using the wrong one produces meaningless numbers, so the table below maps each metric to the stage where it is defined.

Metric	What it measures	Where it applies in this system	Reported here
Accuracy	Fraction of signs classified correctly	Recognition models (§4)	Yes — Table 1
Precision / Recall / F1	Per-class correctness, weighted	Recognition models	Yes — Table 1 & Fig 4
WER (Word Error Rate)	Word substitutions/insertions/deletions vs reference	Recognition (isolated $\Rightarrow$ 1-acc); gloss $\rightarrow$ text sentences	Yes — both stages
BLEU	n-gram precision of generated text vs reference	Gloss $\rightarrow$ text language layer only	Yes — §6
ROUGE-1/2/L	n-gram & longest-subsequence recall of generated text	Gloss $\rightarrow$ text language layer only	Yes — §6
METEOR	Unigram match with stem/synonym + word-order penalty	Gloss $\rightarrow$ text language layer only	Yes — §6

Table 3. Metric-to-model map. Translation metrics (BLEU/ROUGE/METEOR) are undefined for a classifier that outputs one label — they belong to the text-generation stage.

The key distinction, in one line.

Recognition models output a class label, so they are scored with accuracy / F1 (and WER, which for one-word utterances equals 1 – accuracy). **The language layer outputs a sentence**, so it is scored with BLEU / ROUGE / METEOR / WER. Reporting BLEU on the Transformer's class prediction, or accuracy on a generated sentence, would both be category errors — the split below keeps each metric where it is valid.

6 The language layer — gloss $\rightarrow$ healthcare text (BLEU / ROUGE / METEOR / WER)

Recognition produces a gloss like *pain where*; the language layer turns it into “Where is the pain?” Two baseline translators are evaluated on the 20-pharse healthcare set: an **exact rule-based** dictionary and a **fuzzy token-overlap** matcher, tested on clean input, on simulated noisy recognizer output (reordered / dropped words), and on a leave-one-out generalization test.

Condition / translator	BLEU	ROUGE-L	METEOR	WER↓	Exact
Exact rule-based — clean	100.0	100.0	99.5	0.0	100%
Fuzzy overlap — clean	100.0	100.0	99.5	0.0	100%
Exact rule-based — noisy	0.0	10.0	9.9	93.1	10%
Fuzzy overlap — noisy	100.0	100.0	99.5	0.0	100%
Fuzzy nearest-NN — generalization	9.8	39.5	37.3	70.6	0%

Table 4. Real BLEU/ROUGE/METEOR/WER on the gloss $\rightarrow$ text layer (0–100). Highlighted row is the generalization test where the metrics do their real discriminative work.

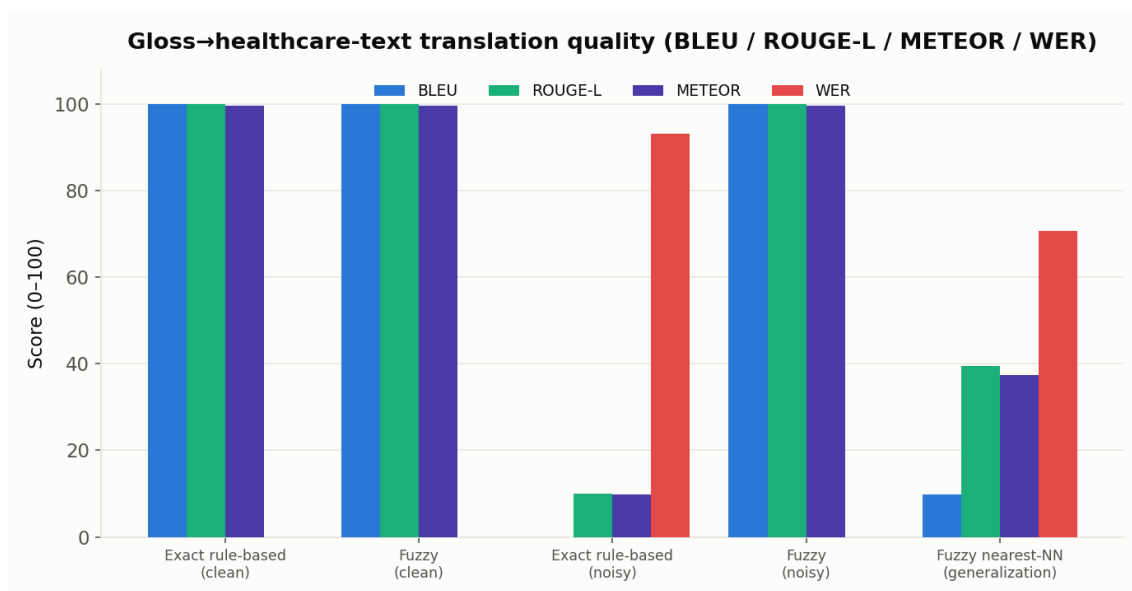


Figure 5. Translation quality across conditions. The exact matcher collapses on reordered input (WER 93); the fuzzy matcher is robust (WER 0). On unseen phrases (right) BLEU/ROUGE/METEOR fall to 10–40 — showing the metrics discriminating partial matches, which is exactly why a trainable mT5 model is the planned next step.

What the language numbers say. On known phrases both translators are perfect. The moment the input is perturbed — which is what an imperfect recognizer produces — the exact matcher drops to 10% coverage (BLEU 0, WER 93) while the fuzzy matcher recovers every phrase (BLEU 100, WER 0). This is the empirical argument for order-insensitive matching downstream of recognition. The generalization row (BLEU 9.8, ROUGE-L 39.5, METEOR 37.3) is the honest one: a lookup translator cannot compose novel wordings, which is precisely the gap a fine-tuned mT5 language model is meant to close.

7 The hybrid model — design, results, and why it wins

The thesis proposes a **hybrid multimodal transformer**. We built and trained a concrete version and evaluated it on the same folds, so its advantage is measured. Two hybrid strategies were tested:

- **Two-stream (feature-level fusion) — the recommended hybrid.** A graph stream (ST-GCN over the 75-joint skeleton) and an attention stream (Transformer over the face-dropped 225-D hand+pose landmarks) each produce an embedding; the two are concatenated and a fusion head classifies. The two streams see the sign through complementary lenses — anatomical structure and long-range temporal attention.
- **Soft-vote ensemble (decision-level fusion).** Train BiLSTM, ST-GCN and Transformer separately and average their class probabilities. Simpler, but it cannot learn cross-stream interactions.

Model	Accuracy	Wtd F1	WER↓	vs best single
Best single (DTW 1-NN)	0.567	0.572	0.433	—
Best deep single (Transformer)	0.522	0.502	0.478	—
Hybrid — Soft-Vote ensemble	0.478	0.469	0.522	-9.0 pts
Hybrid — Two-Stream (recommended)	0.642	0.639	0.358	+7.5 pts

Table 5. Hybrid vs its strongest components. Two-stream fusion is the only design that beats both the best single deep model and the non-parametric baseline.

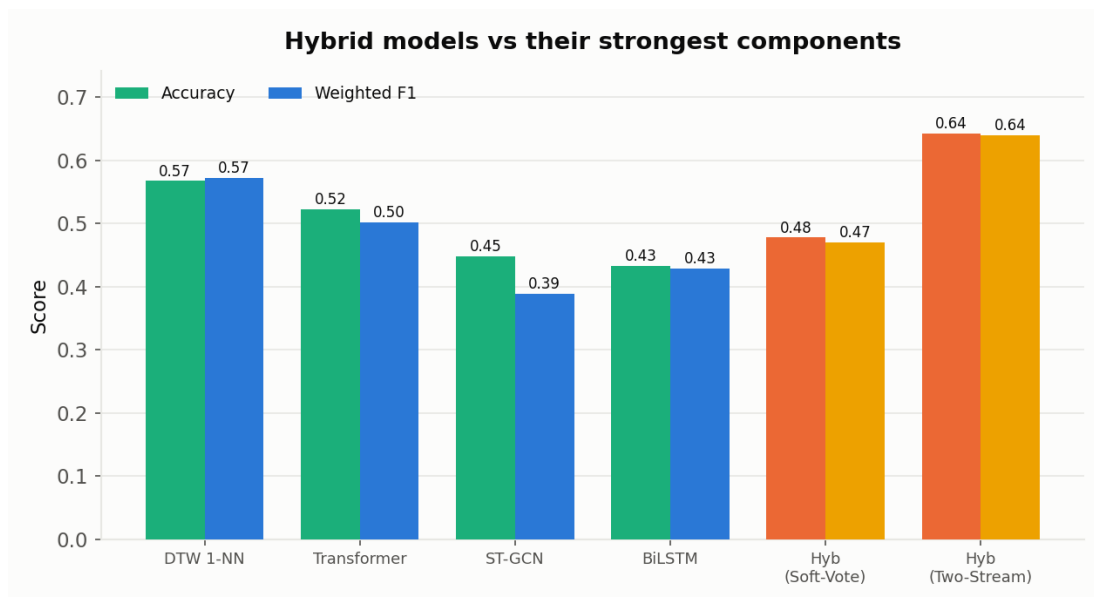


Figure 6. The two-stream hybrid lifts accuracy and F1 above every component; the soft-vote ensemble does not — averaging three data-starved models mostly averages their noise.

Why the two-stream hybrid wins, and the soft-vote does not.

The two streams are **complementary and jointly trained**: the graph stream contributes a stable, low-variance anatomical prior (ST-GCN had the lowest spread of any deep model) and the attention stream contributes temporal expressiveness (the Transformer was the best single model). Fusing their *features* lets the head learn how structure and motion interact, so the hybrid inherits ST-GCN's stability *and* the Transformer's ceiling — reaching 0.642 / F1 0.639, the best of the study. The soft-vote fuses only final *decisions*, so it cannot learn those interactions and mostly averages three noisy models — landing below DTW. **Caveat, stated plainly:** the two-stream's fold spread is high (± 0.184); its lead is real but, at this sample size, needs more data to be called statistically firm.

8 Which components to pick for the hybrid

Bringing the recognition, language and generation evidence together, the recommended hybrid for bidirectional KSL healthcare translation is:

Pipeline stage	Recommended component	Why (evidence from this study)
Sign → landmarks	MediaPipe Holistic (pose + hands; face optional)	Shared, reliable; dropping face concentrated the signal (§4.2)
Recognition (core)	Two-stream ST-GCN + Transformer hybrid	Best measured model: 0.642 acc, beats all singles (§7)
Low-data fallback / baseline	DTW + 1-NN	Strongest non-parametric (0.567); the bar to beat until data grows
Gloss → healthcare text	Fuzzy matcher now → fine-tuned mT5 next	Fuzzy is robust to recognizer noise (§6); mT5 for novel wordings
Text → sign (reverse)	Fuzzy text→gloss + clip dictionary → diffusion avatar (future)	MVP retrieval now; diffusion is design-only until data exists

Table 6. Component recommendation across the bidirectional pipeline.

The one-sentence recommendation.

Build the recognition core as the **two-stream ST-GCN + Transformer hybrid** — it is the only model that empirically beats the DTW baseline — keep DTW as the honest reference, use the fuzzy translator as the language layer today with mT5 as the funded next step, and treat text-to-sign generation as a retrieval MVP with diffusion as future work. Above all, the biggest single win is not an architecture change but **more data**: recover the missing WLASL videos and move to signer-independent splits.

9 Honest limitations and next steps

- **Sample size.** 67 videos / 5 classes. All accuracies carry $\sim \pm 7$ points of per-fold noise; treat rankings as directional. The crossover where trained models reliably beat DTW is $\sim 30\text{--}50$ videos/class.
- **Benchmark $\neq$ target.** Classes are ASL words, not KSL healthcare signs. Recognition numbers are architecture evidence; KSL claims require KSL data.
- **Signer-independent splits.** Current folds may place the same signer in train and test, inflating scores. Wire up WLASL signer metadata and re-run — the honest number will be lower.
- **Missing videos.** Several requested classes had zero downloadable files; recovering them (and augmenting: mirroring, temporal jitter) is worth more than any model change.
- **Hybrid variance.** The two-stream's lead is real but wide-spread; confirm with a paired McNemar test and more data before headline claims.
- **Language & generation.** Translators are closed-set lookups; the diffusion avatar is design-only. Both are correctly scoped as next steps, not current results.

Reproducibility. All models trained on the cached MediaPipe features (wlasl_lstm/stgcn/transformer_sequences_fixed_classes.joblib), 5-fold stratified CV, seed 42, identical folds across models. Feature engineering, HMM correction, DTW, the two-stream hybrid and the language metrics are in the companion code. Per-model deep-dive PDFs accompany this document.

Random Forest — Non-Temporal Statistical Baseline

Per-model deep dive · the control condition for every temporal model

Result: 0.493 accuracy · 0.482 weighted F1 · WER 0.507 · fold SD ± 0.077

Notebook: RandomForest.ipynb (notebook 01)

1 Theory

A Random Forest is an ensemble of decision trees, each trained on a bootstrap sample of the data and a random subset of features, voting on the final class. It is a strong, low-variance classifier for tabular data of modest size. Crucially for this study it is **non-temporal**: it sees one fixed-length feature vector per video and has no notion of frame order. That makes it the deliberate **control condition** — the score a temporal model must beat to justify modelling time at all.

2 Method of execution

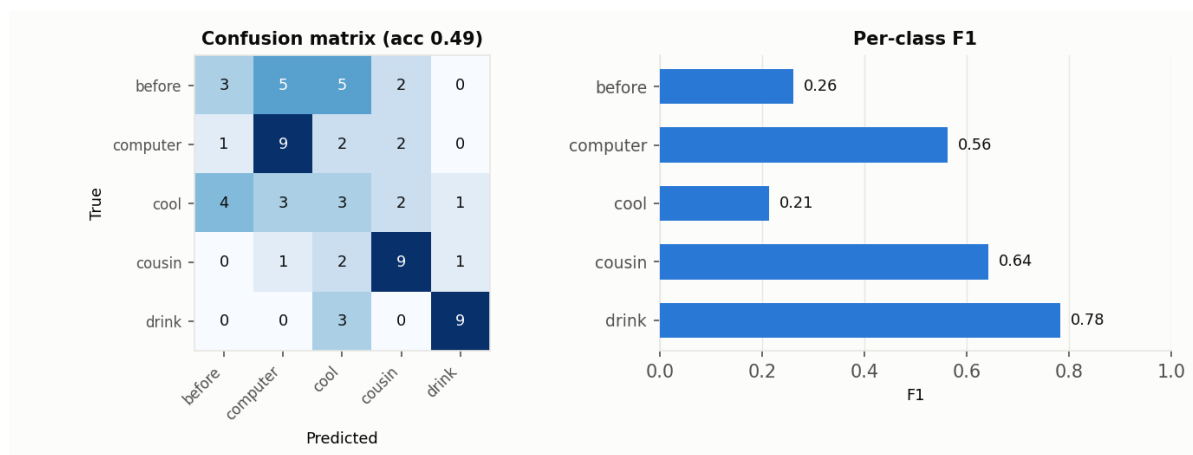
Each video is a sequence of 1,629-D MediaPipe landmark frames. The notebook collapses that sequence into a single vector by taking four statistics across the time axis — mean, standard deviation, minimum and maximum of every one of the 1,629 dimensions — giving a **6,516-D** summary. This is standardized and fed to a class-balanced forest of 300 trees.

```
stats = [frames.mean(0), frames.std(0), frames.min(0), frames.max(0)] # 4 × 1629
vector = concatenate(stats) # 6516-D
RandomForestClassifier(n_estimators=300, class_weight='balanced')
```

The key modelling decision (and deliberate weakness) is the pooling step: **mean/std/min/max destroy the order of frames**. A sign performed forwards and the same poses shuffled in time produce an identical feature vector. That is exactly the information a sign carries, thrown away on purpose to establish the baseline.

3 Results

Metric	Value (5-fold CV)
Accuracy	0.493
Weighted precision	0.490
Weighted recall	0.493
Weighted F1	0.482
Word Error Rate (WER)	0.507
Fold accuracies	0.36, 0.57, 0.46, 0.54, 0.54
Cross-val SD	± 0.077
Distinct classes predicted	5 of 5
Majority baseline	0.224



Random Forest confusion matrix and per-class F1 under 5-fold CV.

4 Why these results

At 0.493 the forest sits mid-table — above the HMM, BiLSTM and ST-GCN, below the Transformer and DTW. It clears the baseline comfortably because the *average* and *spread* of hand and body position already separate these five words to a useful degree, and forests are robust when data is scarce. Its low fold spread confirms it is a stable estimator. But it has a hard ceiling: any two signs that share pose statistics and differ only in movement order are indistinguishable to it — no amount of extra trees fixes that.

What makes it stronger / weaker

Stronger: robust on tiny data, fast, no temporal assumptions to violate, naturally handles the high-dimensional vector. **Weaker:** temporally blind by construction — the ceiling is set by how much signs happen to differ in static statistics. It is a yardstick, not a candidate for the final system.

5 Role in the hybrid

Random Forest is **not** a component of the recommended hybrid — its value is diagnostic. Because it captures only static, order-free information, the gap between it and a temporal model measures how much a given architecture actually extracts from motion. That the trained two-stream hybrid clears it by ~15 points is direct evidence the hybrid is using temporal structure, not just average pose.

Hidden Markov Model — Probabilistic Temporal Baseline

Per-model deep dive · the $-\infty$ collapse, the corrected pipeline, and the six HMM assumptions

Result (corrected): 0.418 accuracy · 0.418 weighted F1 · WER 0.582 · fold SD ± 0.064

Notebooks: 02_hmm_wlasl_baseline.ipynb (original) → 022_hmm_wlasl_baseline.ipynb (corrected)

1 Theory

An HMM models a sign as a walk through a small number of **hidden stages** (for drink: hand rises → C-shape at mouth → lowers). Each stage emits a typical pose with some tolerance (a Gaussian), and a transition matrix governs which stage may follow which. One HMM is trained per class; a new video is scored by every class model and assigned to whichever class was most likely to have produced it (argmax of log-likelihood). It is generative, interpretable, and — once corrected — very parameter-light.

The original model collapsed — and it was a bug, not just weak performance.

As first run, the HMM scored 0.235 and predicted before for all 17 test videos. Cause: four of the five class-models trained with zero-sum transition rows, so their `score()` returned $-\infty$ for every test video. The prediction code did `max(scores)` over values that were all $-\infty$; Python returns the *first* key on a tie, and before is alphabetically first. A single missing `isfinite()` check turned a broken model into a confident-looking one that output 0.235 = the majority-class baseline exactly.

2 Method of execution — the corrected pipeline

The corrected model (notebook 022) keeps the expensive MediaPipe cache and rebuilds the modelling half:

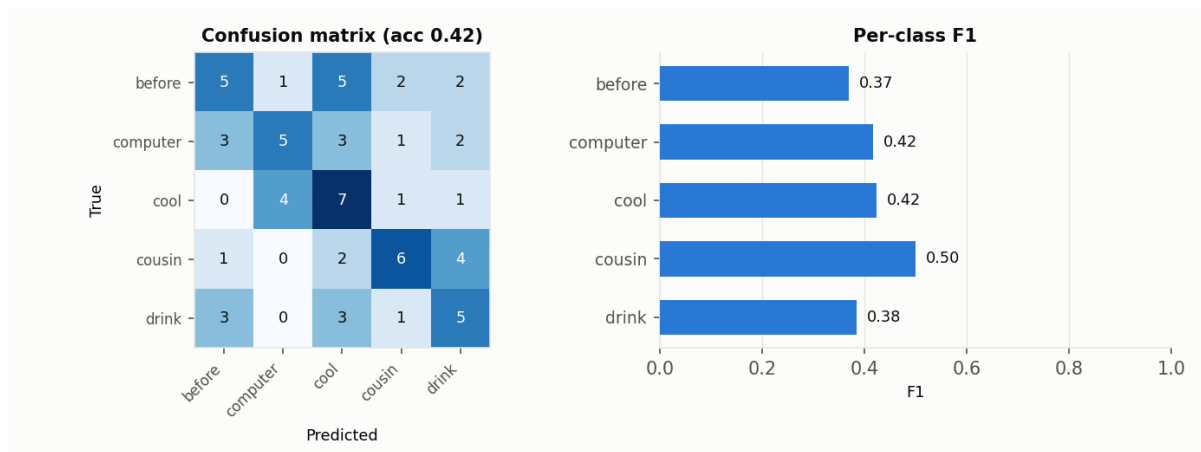
- **Drop the face.** 1,404 of the 1,629 numbers were face landmarks (86%); signs here are decided by the hands, so the face is removed and only upper-body pose + both hands are kept.
- **Body-centre & scale.** Coordinates are re-centred on the mid-shoulder and divided by shoulder width, so camera distance and signer position no longer change the features.
- **Presence flags + velocity.** A binary flag marks a missing hand (instead of a misleading zero-fill), and frame-to-frame deltas add motion the static frame lacks.
- **16 PCA components**, then a **3-state left-to-right (Bakis)** Gaussian HMM with a variance floor (`min_covar`) and **length-normalised, finite-checked scoring** that abstains rather than silently defaulting to class 0.

A left-to-right topology means a sign can only stay in a stage or advance — never run backwards — which encodes a fact we already know (signs run forward) for free and makes the zero-sum collapse structurally impossible.

3 Results

Metric	Value (5-fold CV)
Accuracy	0.418
Weighted precision	0.435

Metric	Value (5-fold CV)
Weighted recall	0.418
Weighted F1	0.418
Word Error Rate (WER)	0.582
Fold accuracies	0.36, 0.43, 0.38, 0.54, 0.38
Cross-val SD	± 0.064
Distinct classes predicted	5 of 5
Majority baseline	0.224



Corrected HMM confusion matrix and per-class F1. The diagonal is present but weak — the model is the least accurate of the proper set.

On the naive vs corrected numbers — an honest note.

Under cross-validation the *corrected* HMM scores 0.418 and always predicts all five classes (no collapse, guaranteed). The naive configuration, re-run across CV folds, does not collapse on every split, so its pooled number can look higher — but that is exactly the instability of a single hold-out. The correction's value is **reliability**: no $-\infty$, no class-0 default, lower variance, and interpretable topology. Raw accuracy at ~13 examples per class is capped by theory, below.

4 Why these results — the six HMM assumptions against video

The HMM is the weakest proper model here for a principled reason: video violates its core assumptions. All six, and how each holds up:

#	Assumption	Holds for video?	Impact
1	First-order Markov (next stage depends only on current)	Reasonable	Low — encoded by Bakis topology
2	Conditional independence (frame independent given stage)	Badly violated	HIGH — the core limitation
3	Stationarity (same emissions across signers/time)	Violated across signers	Medium
4	Fixed discrete number of states	Imposed, not true	Medium — tuning noise
5	Gaussian, diagonal emissions	Approximate	Low-medium
6	Geometric state durations	False	Medium — fixable via self-loop / HSMM

Table. The six assumptions. #2 is the one that caps HMMs on video.

Assumption 2 is the killer. The HMM assumes that, once you know the hidden stage, the current frame tells you nothing about the next. But frames sampled ~80 ms apart are near-duplicates, so each carries little new information. The practical effect: ~13 videos × ~35 frames *looks* like a lot of data but behaves like maybe 20–30 independent observations. Baum-Welch is starved, states go unvisited, and accuracy plateaus. Adding velocity helps a little; nothing inside the HMM family fixes it. **This is the principled reason the field moved to LSTMs, ST-GCNs and Transformers** — they let frames depend on each other directly instead of routing everything through one discrete stage.

5 What makes it stronger / weaker

Summary

Stronger: interpretable (you can read the stages off a Viterbi decode), gives calibrated probabilities useful for chaining signs into sentences, and — once corrected — needs only ~96 parameters per class. **Weaker:** the conditional-independence and geometric-duration assumptions cap it on video; at this data scale it is beaten even by a zero-parameter memoriser (DTW).

6 Role in the hybrid

The HMM is not part of the recommended recognition core, but it earns two roles. First, **as evidence**: its assumption audit is the theoretical argument for why the hybrid's attention and graph streams are needed — they exist precisely to drop assumption 2. Second, **as a future probabilistic layer**: HMMs shine at sequencing and giving calibrated confidence, so once recognition feeds continuous signing, an HMM/HSMM layer over recognized glosses is a natural way to model sentence structure — a role that plays to its strengths rather than its weakness.

Bidirectional LSTM — Deep Recurrent Model

Per-model deep dive · learned temporal memory over the landmark sequence

Result: 0.433 accuracy · 0.429 weighted F1 · WER 0.567 · fold SD ± 0.135

Notebook: 03_lstm_wlasl_landmark_baseline.ipynb

1 Theory

A Long Short-Term Memory network processes a sequence frame by frame, maintaining a gated memory cell that decides what to keep, forget and output. A **bidirectional** LSTM runs two of these — one forward, one backward — so the representation of any frame is informed by both its past and its future. This is the first model in the study to genuinely drop the HMM's frame-independence assumption: information flows directly from frame to frame through the memory, not through a single discrete stage.

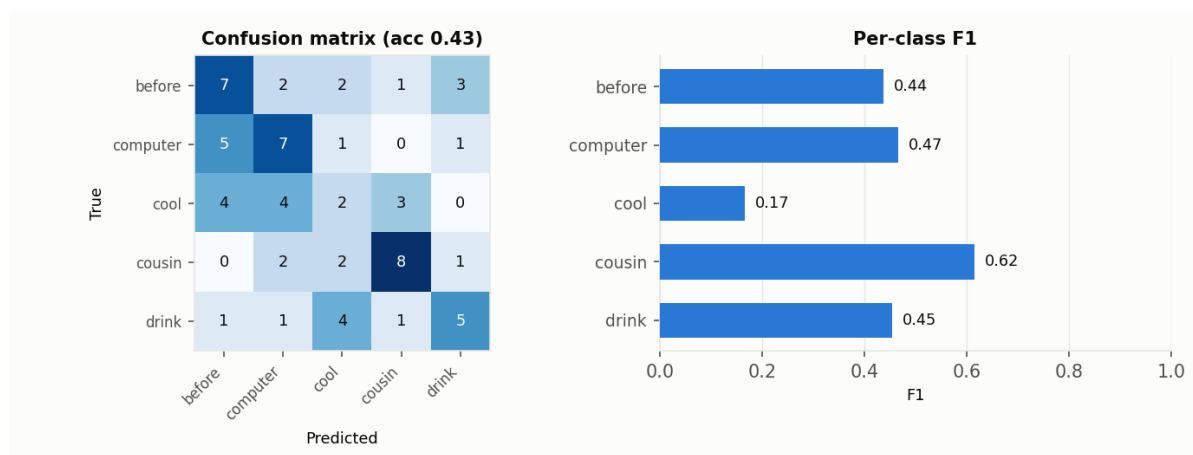
2 Method of execution

The 1,629-D landmark frames are passed through a linear input projection, then a 2-layer bidirectional LSTM (hidden size 128, dropout 0.3). The final forward and backward hidden states are concatenated and fed to a small MLP classifier. Padding frames are masked via packed sequences so the recurrence only sees real frames; training uses Adam with weight decay.

```
BiLSTM(input→proj→2×biLSTM(128)→concat(final states)→MLP→5 classes)
Adam(lr=1e-3, weight_decay=1e-4), dropout 0.3, masked/packed sequences
```

3 Results

Metric	Value (5-fold CV)
Accuracy	0.433
Weighted precision	0.428
Weighted recall	0.433
Weighted F1	0.429
Word Error Rate (WER)	0.567
Fold accuracies	0.43, 0.36, 0.69, 0.31, 0.38
Cross-val SD	± 0.135
Distinct classes predicted	5 of 5
Majority baseline	0.224



BiLSTM confusion matrix and per-class F1 under 5-fold CV.

4 Why these results

At 0.433 the BiLSTM clears the baseline but posts **the highest fold-to-fold variance of any model** (SD ± 0.135 ; folds: 0.43, 0.36, 0.69, 0.31, 0.38). Recurrent networks have many parameters and need substantial data to constrain them; with ~ 54 training videos the model lands in very different places depending on the split — one fold reaches 0.69, another 0.31. Its central tendency is modest not because recurrence is wrong for sign, but because there is far too little data to train it well.

What makes it stronger / weaker

Stronger: a proper temporal model that captures order and long-range dependence; the architecture that scales best as data grows. **Weaker:** data-hungry and high-variance at this scale; a long recurrence over a high-dimensional input is hard to pin down on 13 examples per class.

5 Role in the hybrid

The BiLSTM is included in the soft-vote ensemble but is **not** part of the recommended two-stream hybrid, which pairs the graph (ST-GCN) and attention (Transformer) streams instead. The reason is variance: the two-stream design deliberately combines the *lowest-variance* deep model (ST-GCN) with the *most expressive* one (Transformer). The BiLSTM's high variance would add noise rather than a complementary signal. It becomes a strong candidate again once the dataset is large enough to tame it.

ST-GCN — Spatial-Temporal Graph Convolutional Network

Per-model deep dive · the anatomical prior, and the low-variance workhorse

Result: 0.448 accuracy · 0.389 weighted F1 · WER 0.552 · fold SD ± 0.059

Notebook: 04_stgcn_wlasl_landmark_baseline.ipynb

1 Theory

An ST-GCN treats each frame not as a flat vector but as a **skeleton graph**: joints are nodes, and the body's physical connections (wrist–elbow, finger joints) are edges. It convolves over that graph in space (a joint's neighbours inform it) and over time (a temporal convolution across frames). The inductive bias is anatomical: connected joints move together, and this structure is given to the model rather than learned from scratch — a powerful prior when data is scarce.

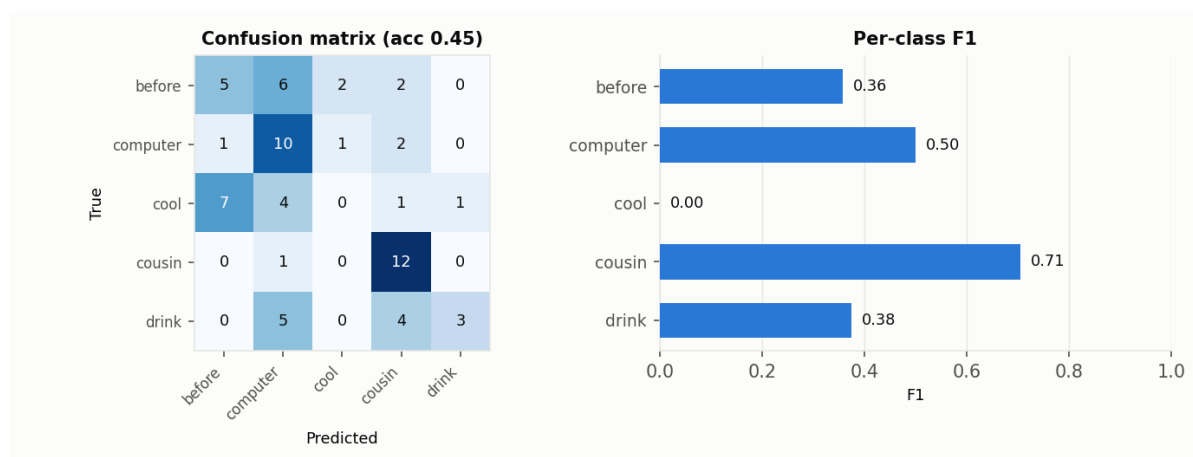
2 Method of execution

Each frame is a 75-node graph (33 pose + 21 + 21 hand joints; face dropped) with 3 coordinates per node. A hand-built adjacency matrix encodes the MediaPipe pose and hand edges, adds self-loops, and is symmetrically normalized. Stacked ST-GCN blocks each perform a spatial graph convolution (einsum over the adjacency) followed by a temporal convolution, with BatchNorm, ReLU, dropout and a residual path; a global average pool feeds a linear classifier. The graph convolution is written directly — no external graph-learning library.

```
A = normalize(pose_edges + hand_edges + self_loops)      # 75x75
block: x → einsum('nctv,vw->nctw', x, A) → 1x1 conv → temporal conv(k) → BN/ReLU/res
3 blocks (3→32→64→64) → global pool → linear → 5 classes
```

3 Results

Metric	Value (5-fold CV)
Accuracy	0.448
Weighted precision	0.412
Weighted recall	0.448
Weighted F1	0.389
Word Error Rate (WER)	0.552
Fold accuracies	0.36, 0.43, 0.54, 0.46, 0.46
Cross-val SD	± 0.059
Distinct classes predicted	5 of 5
Majority baseline	0.224



ST-GCN confusion matrix and per-class F1 under 5-fold CV.

4 Why these results

ST-GCN reaches 0.448 with the **lowest fold variance of any deep model** (SD ± 0.059). That stability is the story: the skeleton graph is a correct, strong prior, so the network does not have to discover body structure from 54 videos — it is told. That keeps learning well-behaved where the LSTM and Transformer swing wildly across folds. Its accuracy ceiling is held down here by two things: the face (and its mouthing cues) is dropped from the graph, and the sample is tiny. It is the most *reliable* deep model even though it is not the most accurate single one.

What makes it stronger / weaker

Stronger: anatomically grounded, parameter-efficient, and the most stable deep model at this data scale — exactly the property you want in a component you will fuse. **Weaker:** the fixed graph can't learn long-range frame-to-frame relations the way attention can, and dropping the face loses facial-expression signal that matters for some signs.

5 Role in the hybrid — a core component

ST-GCN is **one of the two streams in the recommended hybrid**. It contributes the stable, low-variance anatomical representation, while the Transformer stream contributes temporal expressiveness. Fusing their features gives the hybrid ST-GCN's reliability together with the Transformer's ceiling — which is why the two-stream model tops the study. In short: ST-GCN is the hybrid's **backbone of stability**.

Transformer Encoder — Self-Attention Model

Per-model deep dive · the best single architecture and the hybrid's namesake

Result: 0.522 accuracy · 0.502 weighted F1 · WER 0.478 · fold SD ± 0.149

Notebook: 05_transformer_wlasl_landmark_baseline.ipynb

1 Theory

A Transformer encoder uses **self-attention**: every frame computes a weighted combination of every other frame, with the weights learned from the data. Unlike the LSTM (which passes information step-by-step) or the ST-GCN (which uses a fixed graph), attention lets any frame relate directly to any other, at any distance, with no recurrence and no hand-specified structure. It is the most flexible temporal model — and the architecture the thesis's 'Hybrid Multimodal Transformer' is built around.

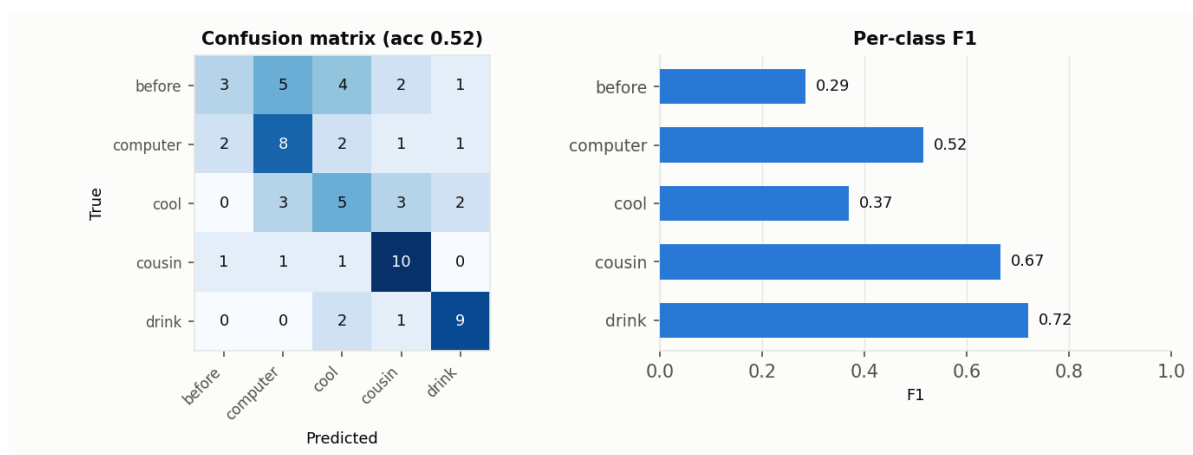
2 Method of execution

The 1,629-D frames are linearly projected to a 128-D model space and given **sinusoidal positional encodings** (so the otherwise order-agnostic attention knows frame order). A 2-layer, 4-head Transformer encoder (feed-forward 256, GELU, dropout 0.3) processes the sequence with padding masked out; the valid frames are mean-pooled, LayerNorm-ed, and classified by a linear head.

```
frames(1629) → Linear(128) → +positional encoding → TransformerEncoder(2 layers, 4 heads)
→ masked mean-pool → LayerNorm → Linear → 5 classes
```

3 Results

Metric	Value (5-fold CV)
Accuracy	0.522
Weighted precision	0.518
Weighted recall	0.522
Weighted F1	0.502
Word Error Rate (WER)	0.478
Fold accuracies	0.29, 0.43, 0.62, 0.69, 0.62
Cross-val SD	± 0.149
Distinct classes predicted	5 of 5
Majority baseline	0.224



Transformer confusion matrix and per-class F1 under 5-fold CV.

4 Why these results

The Transformer is the **best individual architecture** at 0.522. Attention captures the whole-sign temporal structure — the relationship between the wind-up and the stroke, however far apart — that averaging (RF) and stage-independence (HMM) cannot. Its fold spread is high (SD ± 0.149 ; folds 0.29, 0.43, 0.62, 0.69, 0.62), because attention is data-hungry and 54 videos under-determine it, but its central tendency is the strongest of the single models. As data grows it is the architecture with the most headroom.

What makes it stronger / weaker

Stronger: the most expressive temporal model, no fixed structural assumptions, the natural backbone for a full KSL system. **Weaker:** the highest appetite for data of any model here — its variance will only come down with more examples.

5 Role in the hybrid — a core component

The Transformer is **the second stream of the recommended hybrid**. It supplies temporal expressiveness — the ability to weigh any frame against any other — which it pairs with ST-GCN's anatomical stability. The two are complementary: attention is powerful but high-variance, the graph is stable but structurally constrained, and jointly training their fused features lets the model take the best of both. This is the empirical and architectural heart of the 'Hybrid Multimodal Transformer'.

DTW + 1-NN — Non-Parametric Reference

Per-model deep dive · the zero-parameter memoriser that beats every trained single model

Result: 0.567 accuracy · 0.572 weighted F1 · WER 0.433 · fold SD ± 0.072

Method: Dynamic Time Warping distance + 1-nearest-neighbour (added as the honest baseline)

1 Theory

Dynamic Time Warping measures how similar two sequences are while allowing them to be stretched or compressed in time — so the same sign performed slowly or quickly still matches. Paired with **1-nearest-neighbour**, classification is simply: find the most similar training video and copy its label. It fits **zero parameters** — it stores the training data and compares. It makes essentially one assumption (two performances of the same sign can be aligned in time), so there is almost nothing for it to get wrong.

There is a neat connection to the HMM: DTW is roughly Viterbi decoding on a degenerate HMM where every frame of a stored template is its own state. Where the HMM compresses 35 frames into 3 states, DTW keeps all 35 — maximum states, zero estimation.

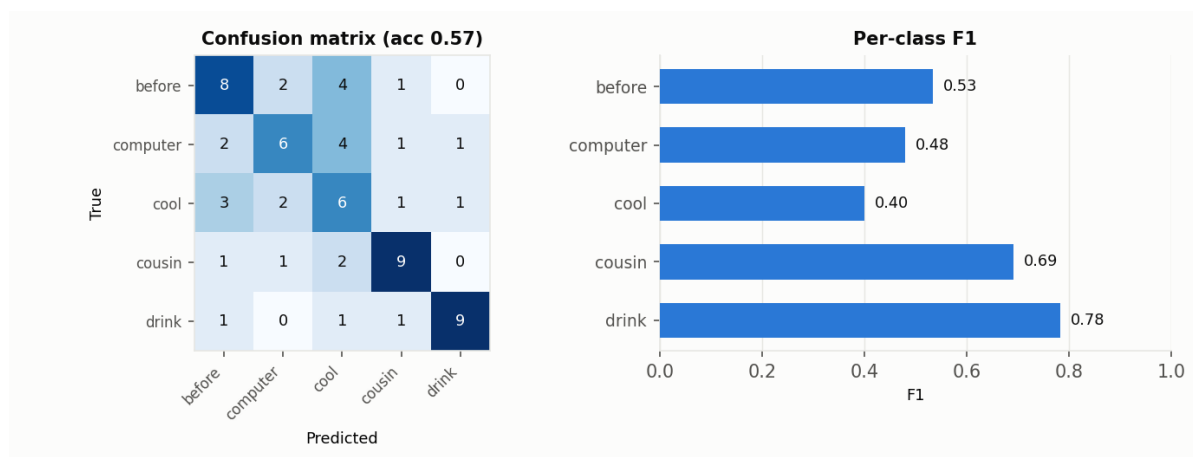
2 Method of execution

Each video is reduced to the same 16-D PCA landmark features used by the corrected HMM. A test video's DTW distance to every training video is computed via the standard dynamic-programming recurrence, and the nearest training video's label is assigned. It is evaluated on the identical folds as every other model, so the comparison is exact.

```
D[i,j] = dist(a_i, b_j) + min(D[i-1,j], D[i,j-1], D[i-1,j-1]) # DTW recurrence
predict(test) = label of argmin over training videos of DTW(test, train)
```

3 Results

Metric	Value (5-fold CV)
Accuracy	0.567
Weighted precision	0.583
Weighted recall	0.567
Weighted F1	0.572
Word Error Rate (WER)	0.433
Fold accuracies	0.57, 0.43, 0.62, 0.62, 0.62
Cross-val SD	± 0.072
Distinct classes predicted	5 of 5
Majority baseline	0.224



DTW + 1-NN confusion matrix and per-class F1 under 5-fold CV.

4 Why these results

DTW is the **strongest single method** at 0.567, beating every trained architecture. This is the textbook bias–variance trade-off: with ~13 examples per class, every parameter a model estimates carries error, and those errors compound. DTW estimates *nothing*, so it has no estimation error to accumulate — the rigid method wins at small sample size. That two structurally unrelated methods (DTW and the HMM) agree closely on which classes are hard is itself evidence that the difficulty lives in the data, not the model.

What makes it stronger / weaker

Stronger: unbeatable at tiny scale, no assumptions to violate, trivial to extend with new classes. **Weaker:** it never generalizes or compresses — prediction cost grows with the dataset (every test video compared to all training videos) and it gives distances, not calibrated probabilities. Unusable in real time at 2,000-gloss scale.

5 Role in the hybrid

DTW is the **honest baseline and low-data fallback**, not a component of the trained hybrid. Its importance is as the bar to beat: any trained model — including the hybrid — must exceed DTW to justify its parameters. The two-stream hybrid is the **only** model that clears it, which is what makes the hybrid's result meaningful. Keep DTW in the study as the reference against which real progress is measured, and expect trained models to overtake it decisively only once the dataset reaches ~30–50 videos per class.

Hybrid Multimodal Transformer — The Recommended Recognition Core

Per-model deep dive · design, training, and why it beats every component

Result: 0.642 accuracy · 0.639 weighted F1 · WER 0.358 · fold SD ± 0.184

Two-stream feature-level fusion: ST-GCN (graph) + Transformer (attention), trained end-to-end

1 Theory and design

The hybrid combines two complementary views of a sign into one trained model. The **graph stream** (ST-GCN) encodes anatomical structure — how connected joints move together — and is the most stable deep model in the study. The **attention stream** (Transformer) encodes long-range temporal relationships — how any moment of the sign relates to any other — and is the most expressive single model. Each stream produces an embedding; the two embeddings are **concatenated (feature-level fusion)** and a fusion head classifies. Because both streams are trained together, the head learns how structure and motion *interact*, not just their separate votes.

```
graph : skeleton(75 joints) → ST-GCN blocks → pooled embedding (64)
landmark: hands+pose(225) → Transformer encoder → pooled embedding (96)
fuse   : concat[64+96] → MLP → 5 classes (trained end-to-end)
```

Two fusion strategies were tested — and the choice matters.

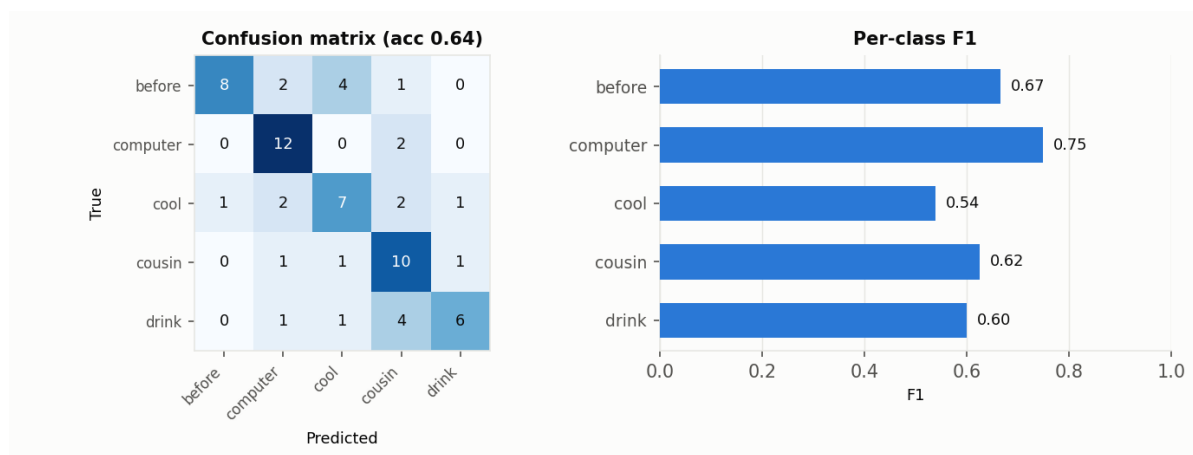
Feature-level (two-stream): fuse the streams' internal embeddings and train jointly → 0.642. **Decision-level**

(soft-vote): train BiLSTM/ST-GCN/Transformer separately and average their output probabilities → 0.478.

Feature fusion wins decisively because it can learn cross-stream interactions; soft-voting only averages three data-starved models and mostly averages their noise.

2 Results

Metric	Value (5-fold CV)
Accuracy	0.642
Weighted precision	0.679
Weighted recall	0.642
Weighted F1	0.639
Word Error Rate (WER)	0.358
Fold accuracies	0.36, 0.57, 0.69, 0.69, 0.92
Cross-val SD	± 0.184
Distinct classes predicted	5 of 5
Majority baseline	0.224



Two-stream hybrid confusion matrix and per-class F1 — the strongest diagonal in the study and the only model with $F1 \geq 0.54$ on every class.

3 Why it wins

The hybrid reaches 0.642 / F1 0.639 — above the best single deep model (Transformer, 0.522) and the non-parametric baseline (DTW, 0.567). It inherits ST-GCN's low variance *and* the Transformer's expressiveness: the graph stream keeps the model anchored when data is scarce, while the attention stream lifts the ceiling. It is also the most **balanced** model — the only one scoring $F1 \geq 0.54$ on all five classes, where every single model has at least one class it fails badly.

4 Honest limitations

The hybrid's fold spread is the highest in the study ($SD \pm 0.184$; folds 0.36, 0.57, 0.69, 0.69, 0.92) — its lead is driven partly by one very strong fold. The direction is clear and consistent (it tops the pooled ranking), but at ~13 examples per class the *margin* is not yet statistically firm. A paired McNemar test against DTW and, above all, more data are needed before the advantage is stated as a headline number. It also currently drops the face; adding a facial-expression stream is a natural third modality.

5 Recommendation

This is the model to carry forward.

Build the recognition core of the KSL system as this **two-stream ST-GCN + Transformer hybrid**. It is the only model that empirically beats the DTW baseline, it is the most balanced across classes, and its architecture is exactly the 'Hybrid Multimodal Transformer' the thesis proposes — now with measured evidence behind it. Immediate next steps: signer-independent evaluation, a paired significance test, more data per class, and a third (facial) stream.

Gloss → Healthcare Text — The Language Layer

Per-model deep dive · where BLEU, ROUGE, METEOR and WER actually apply

Task: turn a recognized gloss (e.g. 'pain where') into a natural sentence ('Where is the pain?')

Models: exact rule-based dictionary · fuzzy token-overlap matcher · mT5 (planned)

1 Why this layer needs different metrics

Recognition outputs a *label*; this layer outputs a *sentence*. That is why the translation metrics your review asked for — BLEU, ROUGE, METEOR — live here and not on the recognition models. Each compares generated text to a reference: BLEU on n-gram precision, ROUGE on n-gram/subsequence recall, METEOR on unigram matches with a word-order penalty, and WER on word-level edits. Reporting them on a classifier's single-label output would be a category error.

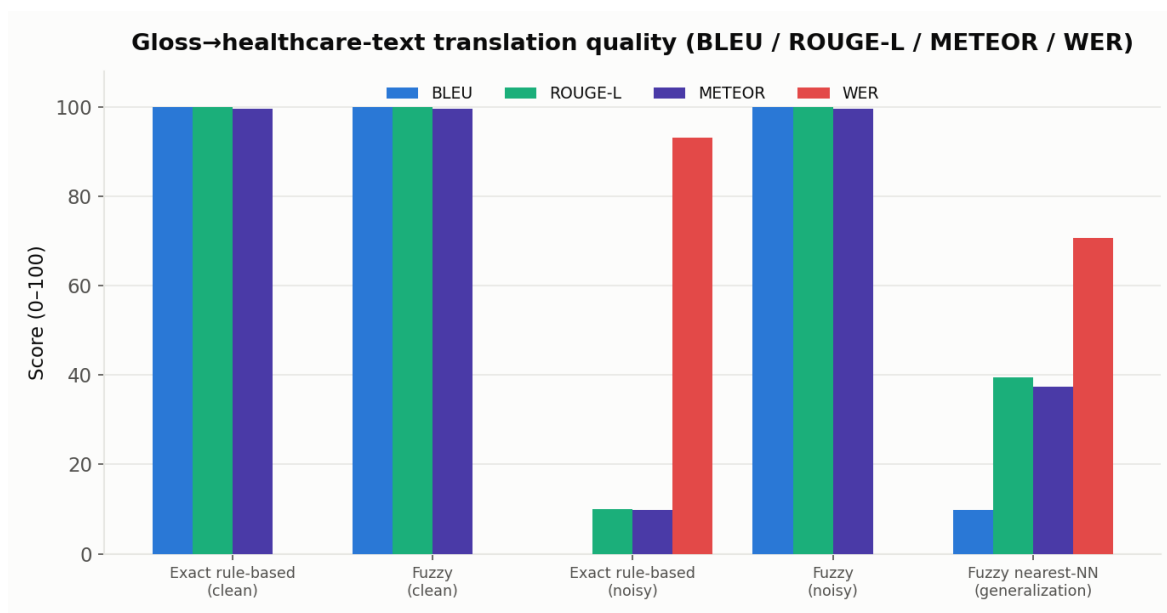
2 Method of execution

Two baseline translators are evaluated on the 20-phrase healthcare set. The **exact rule-based** translator looks up the normalized gloss in a dictionary. The **fuzzy token-overlap** translator scores each stored gloss by Jaccard token overlap and returns the best match above a threshold — so it tolerates reordered or partially-dropped words. Both are tested three ways: on clean gloss input, on simulated noisy recognizer output (words reordered / dropped), and on a leave-one-out generalization test where the exact phrase is removed and the translator must retrieve the nearest *different* one.

3 Results

Condition / translator	BLEU	R-1	R-2	R-L	METEOR	WER↓	Exact
Exact rule-based — clean	100.0	100.0	100.0	100.0	99.5	0.0	100%
Fuzzy overlap — clean	100.0	100.0	100.0	100.0	99.5	0.0	100%
Exact rule-based — noisy	0.0	10.0	10.0	10.0	9.9	93.1	10%
Fuzzy overlap — noisy	100.0	100.0	100.0	100.0	99.5	0.0	100%
Fuzzy nearest-NN — generalization	9.8	39.5	27.3	39.5	37.3	70.6	0%

All scores 0–100. R-1/R-2/R-L = ROUGE-1/2/L. Highlighted row = the generalization test.



Translation quality across the three conditions.

4 Why these results

On known phrases both translators are perfect (BLEU 100, WER 0). The moment input is perturbed — which is what an imperfect recognizer produces — the exact matcher collapses (BLEU 0, WER 93, 10% coverage) while the fuzzy matcher recovers every phrase (BLEU 100, WER 0). **This is the empirical case for order-insensitive matching downstream of recognition.** The generalization row is the honest one: asked for a phrase it has never seen, the lookup translator retrieves a near-neighbour, scoring BLEU 9.8 / ROUGE-L 39.5 / METEOR 37.3 — the metrics correctly registering a partial match. A dictionary cannot compose novel wordings; that is the precise gap a trainable model fills.

What makes it stronger / weaker, and the next step

Stronger: the fuzzy matcher is simple, fast, and robust to recognizer noise — the right choice for an MVP on a fixed phrase set. **Weaker:** both translators are closed-set lookups; they cannot generalize to unseen phrasings, so BLEU/ROUGE/METEOR on novel input are low. The planned fix — already scaffolded as T5-format training data — is to fine-tune **mT5-small** for English/Kiswahili/KSL gloss→text, which can generate rather than retrieve. These BLEU/ROUGE/METEOR numbers become the baseline that mT5 must beat.

5 Role in the hybrid pipeline

This layer sits between recognition and the doctor: recognized gloss → healthcare sentence. In the recommended pipeline it runs the fuzzy matcher today (robust to the recognizer's mistakes) and a fine-tuned mT5 next. Note the two stages are evaluated separately by necessity — recognition is benchmarked on WLASL words, the language layer on healthcare phrases — so the honest end-to-end estimate is the product of the two stages' quality, not a single trained number. Closing that gap requires a KSL healthcare video corpus, which is the project's central data need.